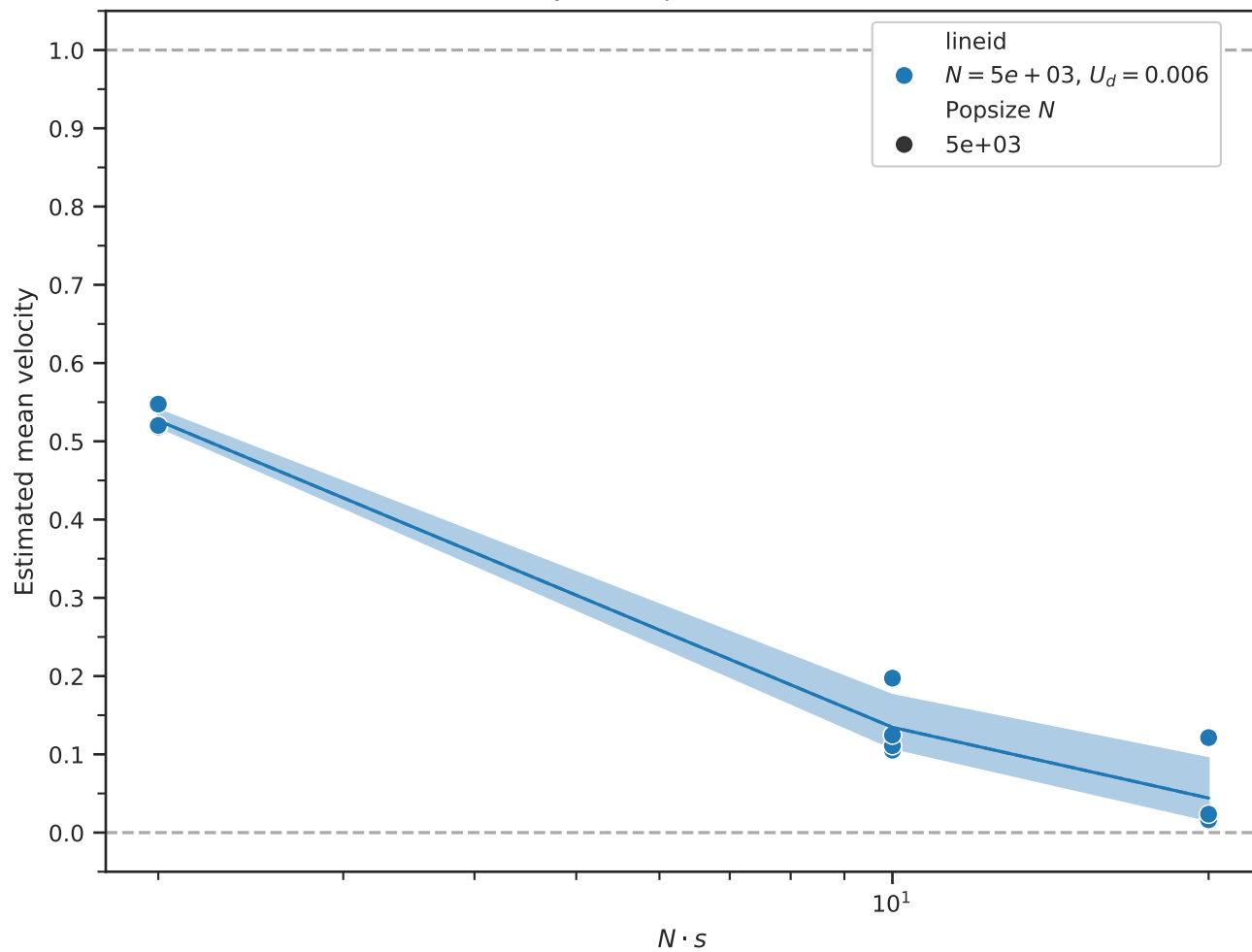
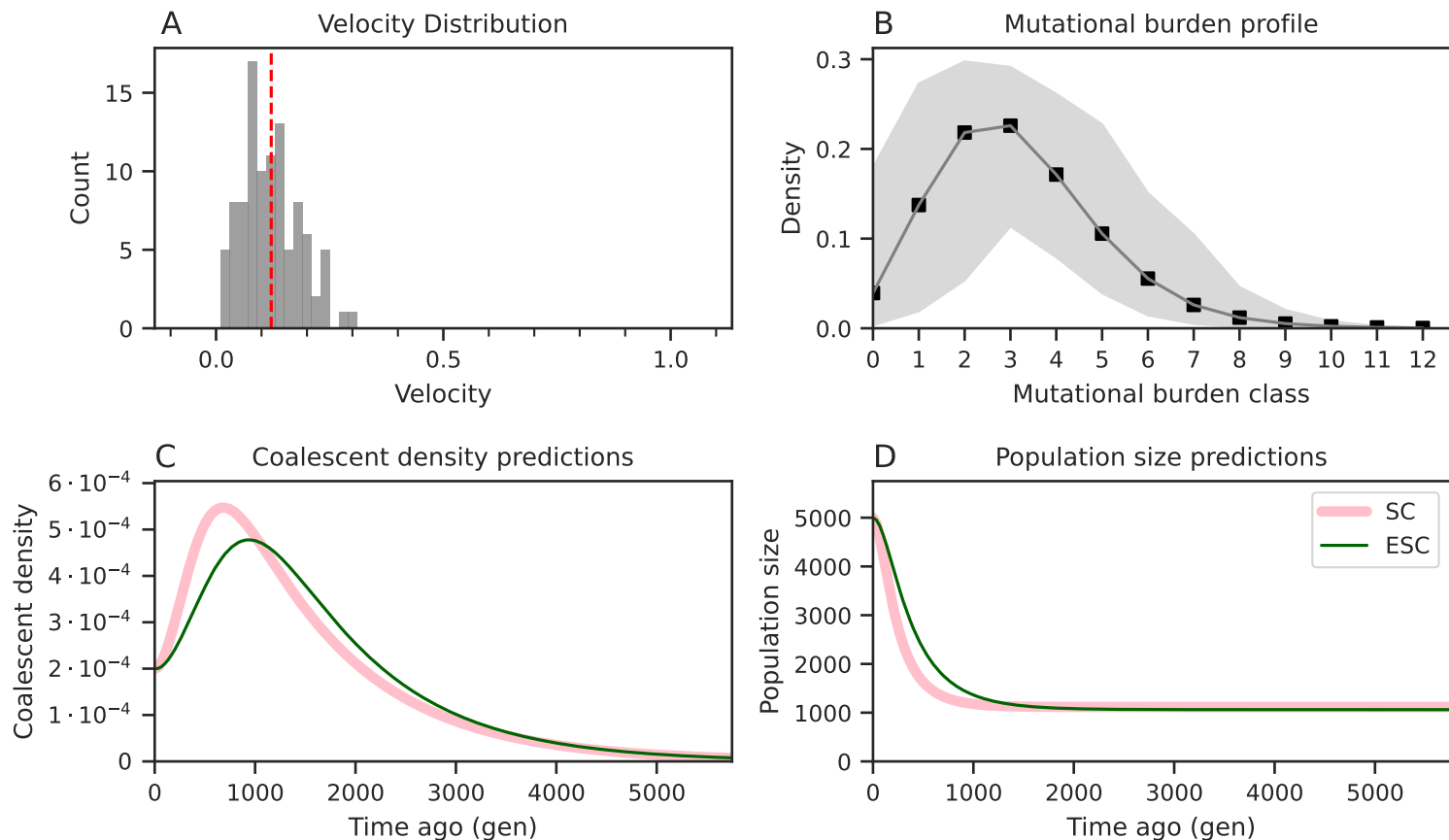


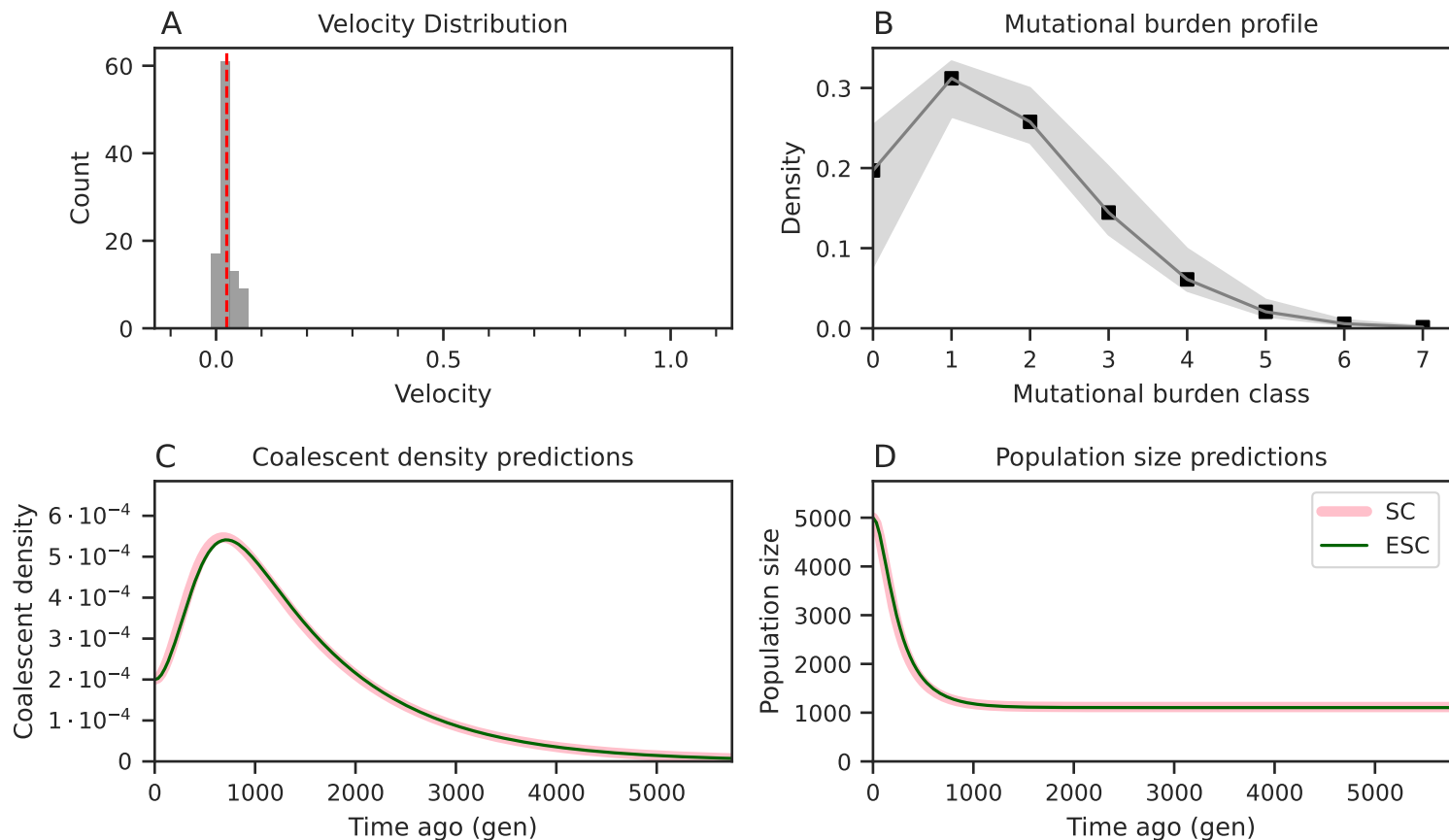
Mean velocity across parameter combinations



$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.002$, $\phi = 4.46$
Mean Velocity= 0.1213 ± 0.0616 , $n = 100$ simulations

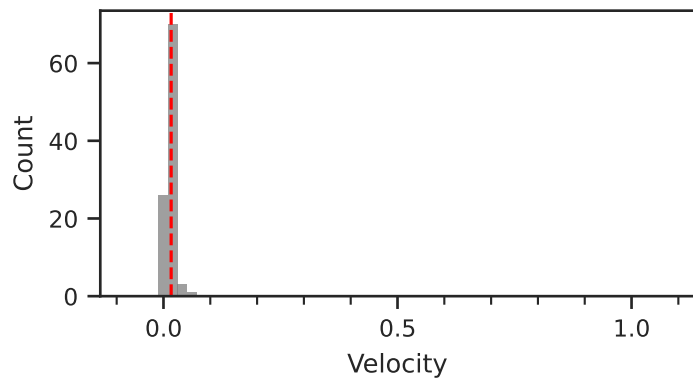


$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 0.001$, $\phi = 4.46$
Mean Velocity= 0.0235 ± 0.0152 , $n = 100$ simulations

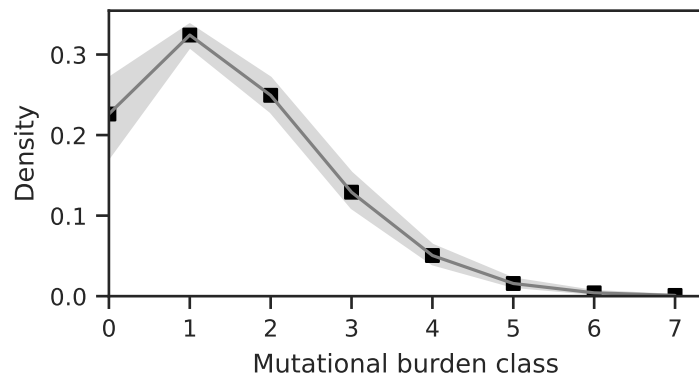


$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 4.46$
Mean Velocity= 0.0164 ± 0.0091 , $n = 100$ simulations

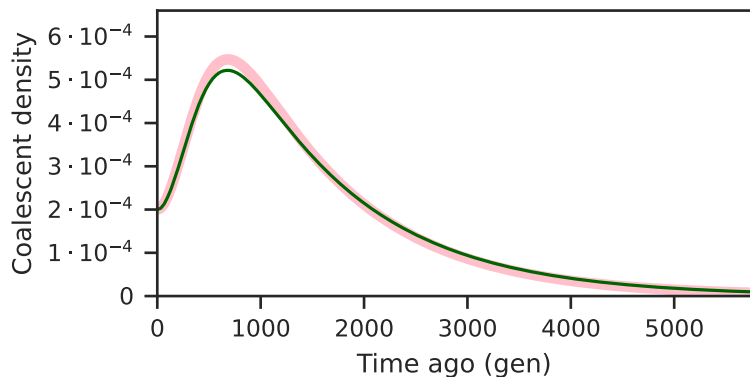
A Velocity Distribution



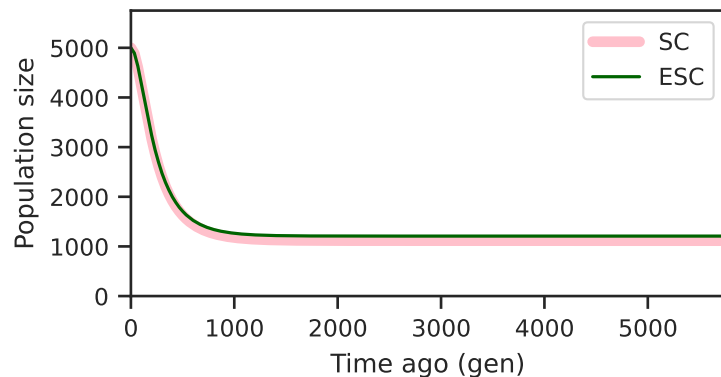
B Mutational burden profile



C Coalescent density predictions

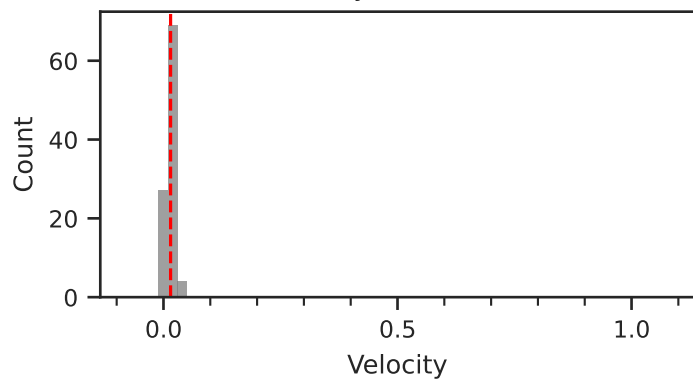


D Population size predictions

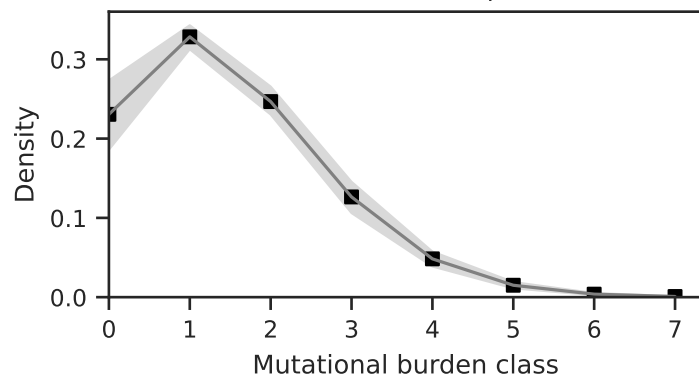


$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-4}$, $\phi = 4.46$
Mean Velocity= 0.0152 ± 0.0086 , $n = 100$ simulations

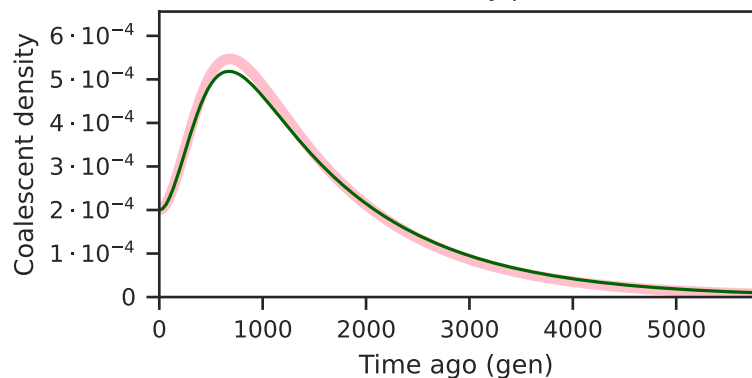
A Velocity Distribution



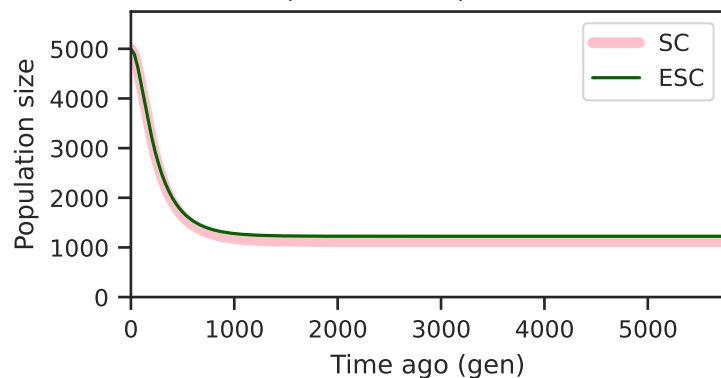
B Mutational burden profile



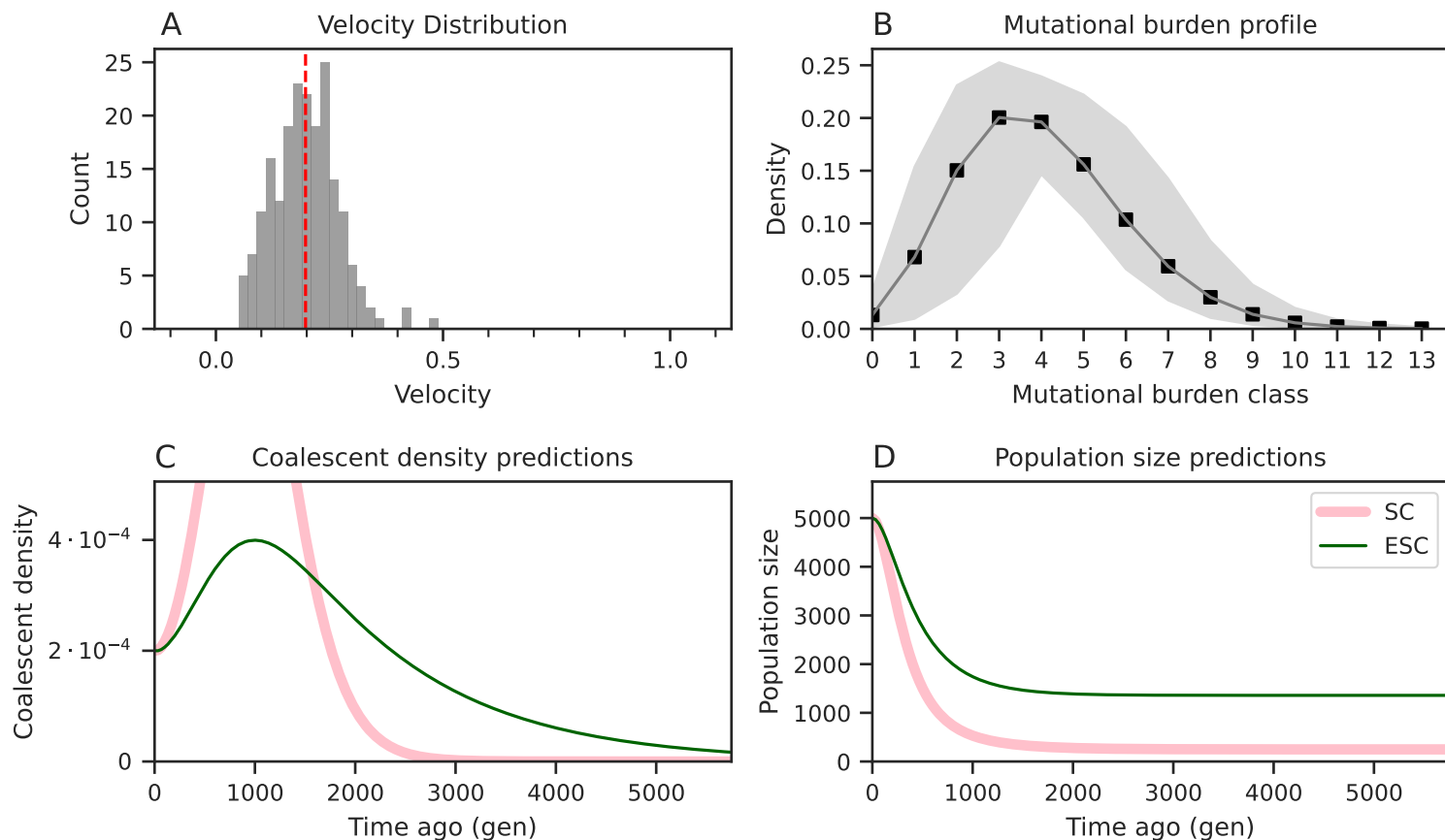
C Coalescent density predictions



D Population size predictions

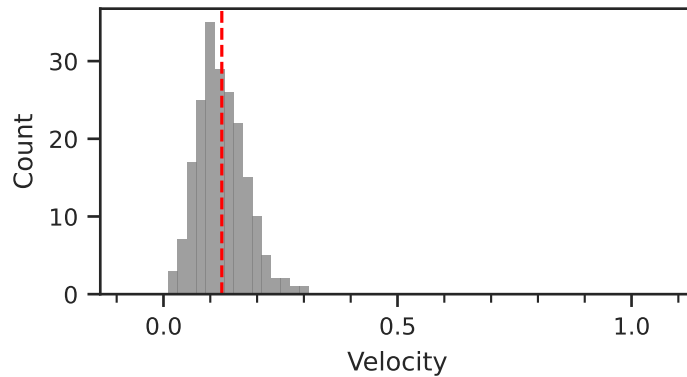


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 0.001$, $\phi = 0.498$
Mean Velocity= 0.1974 ± 0.0715 , $n = 200$ simulations

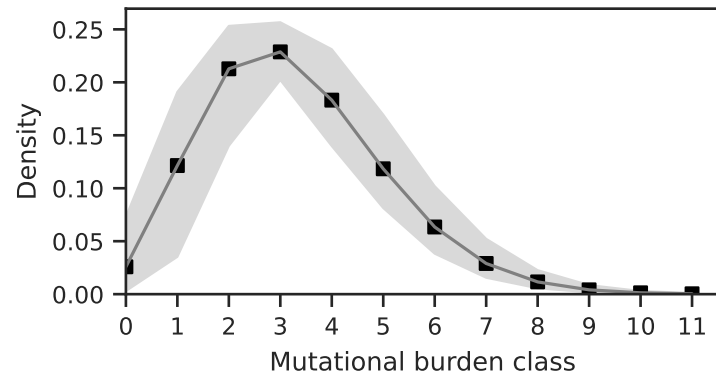


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity = 0.1246 ± 0.0503 , $n = 200$ simulations

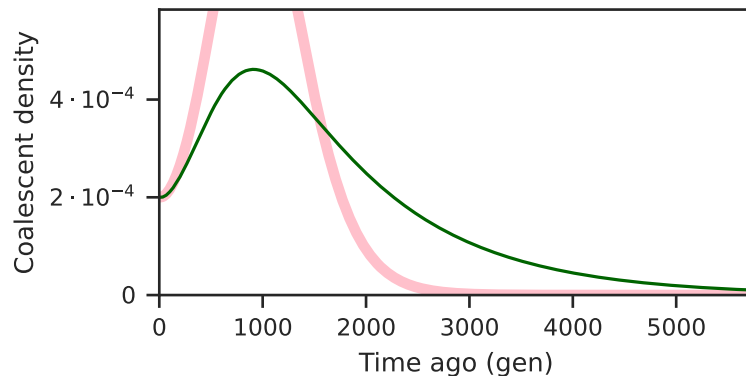
A Velocity Distribution



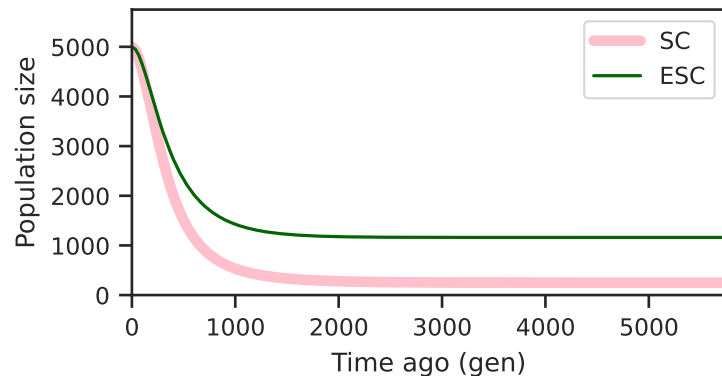
B Mutational burden profile



C Coalescent density predictions

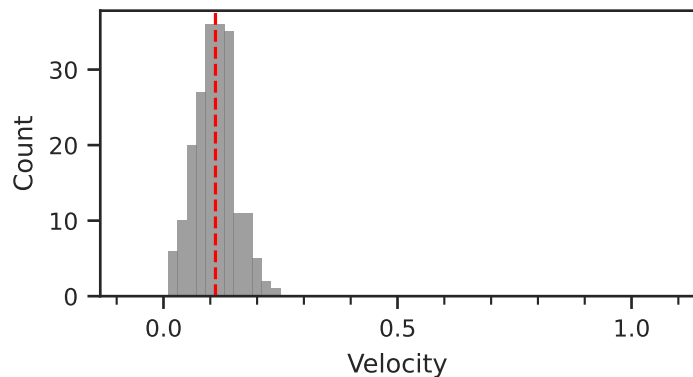


D Population size predictions

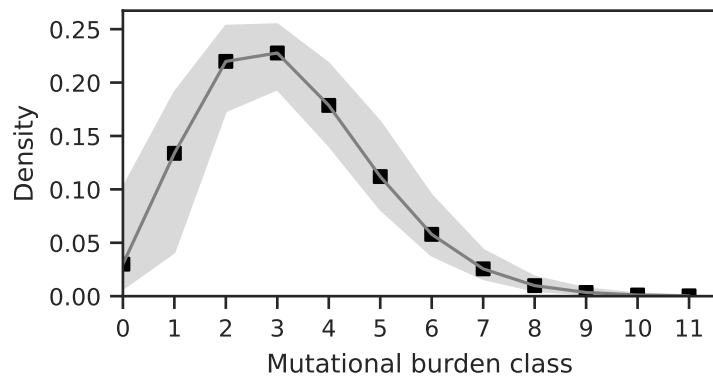


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity= 0.1111 ± 0.0432 , $n = 200$ simulations

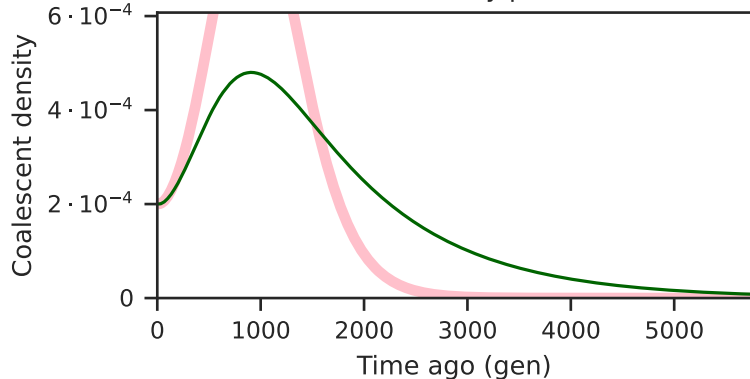
A Velocity Distribution



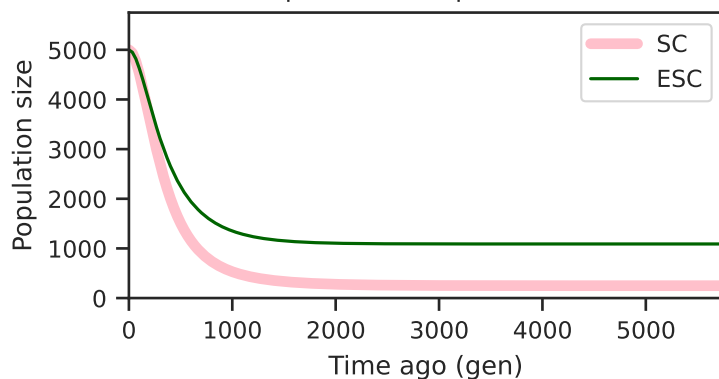
B Mutational burden profile



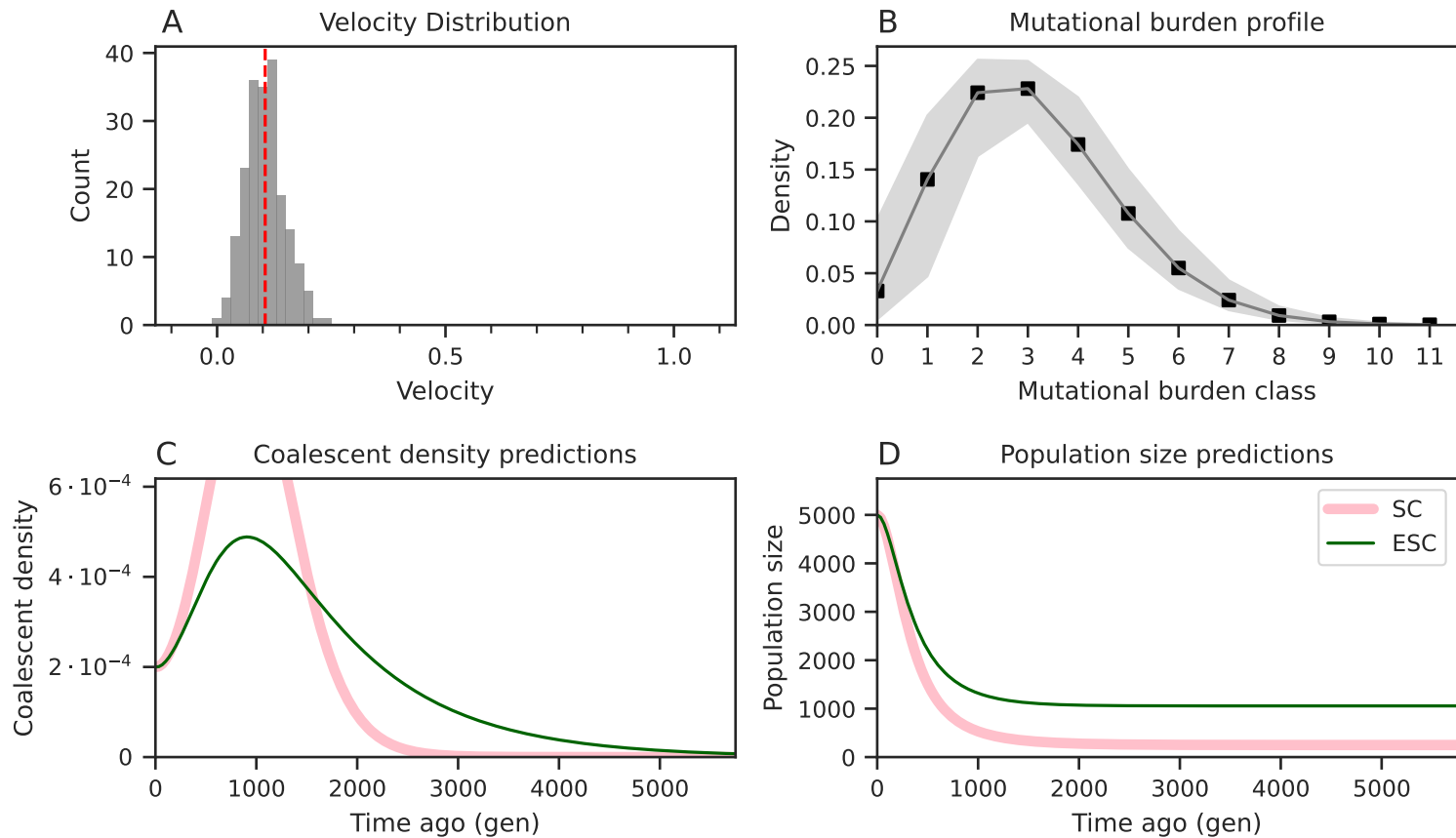
C Coalescent density predictions



D Population size predictions

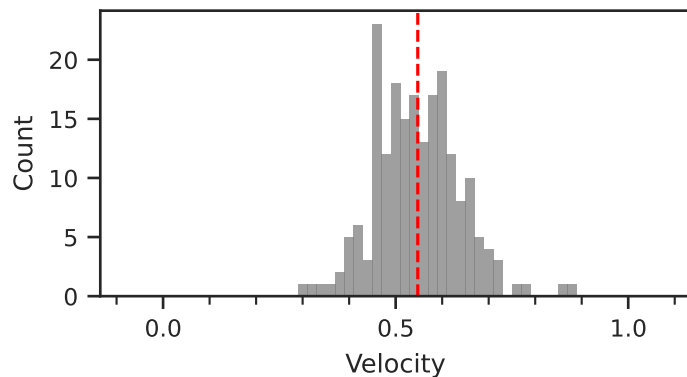


$N = 5e + 03$, $s = 0.002$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 0.498$
Mean Velocity = 0.1053 ± 0.0424 , $n = 200$ simulations

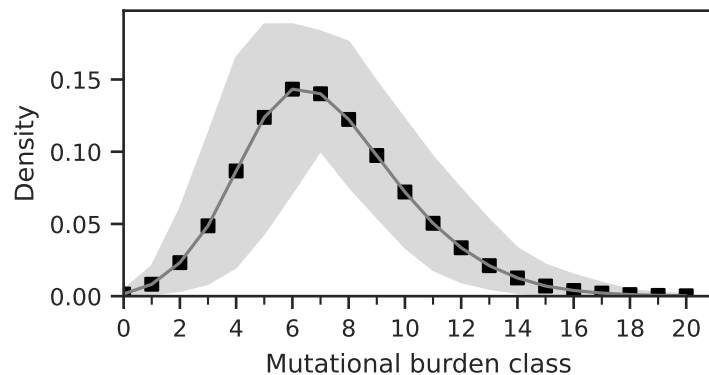


$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 2 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5475 ± 0.0915 , $n = 200$ simulations

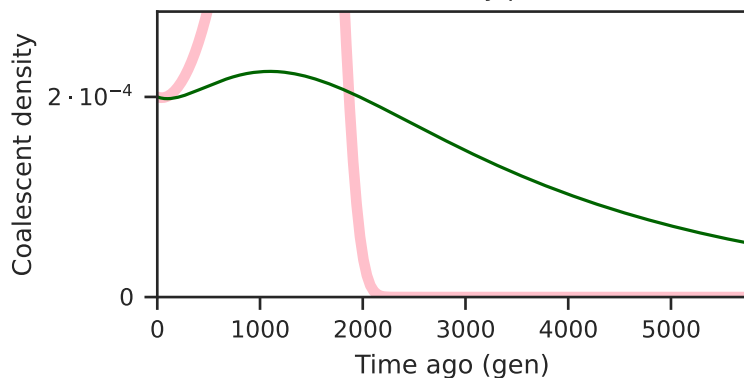
A Velocity Distribution



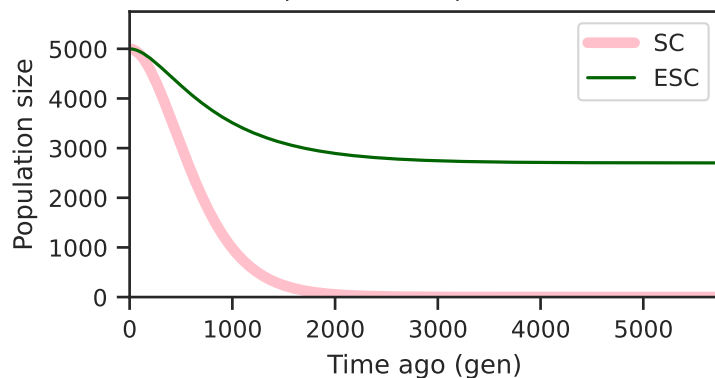
B Mutational burden profile



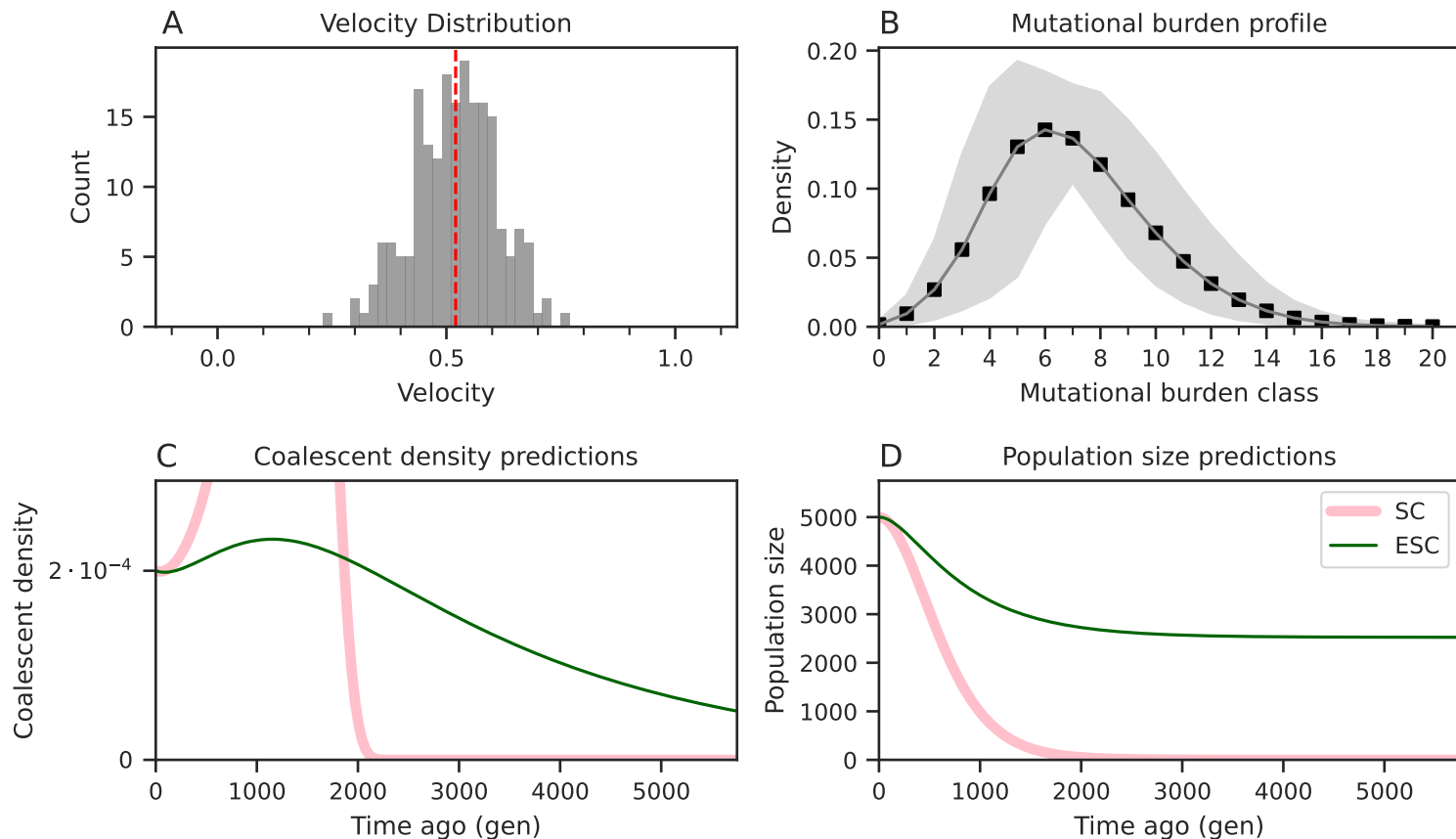
C Coalescent density predictions



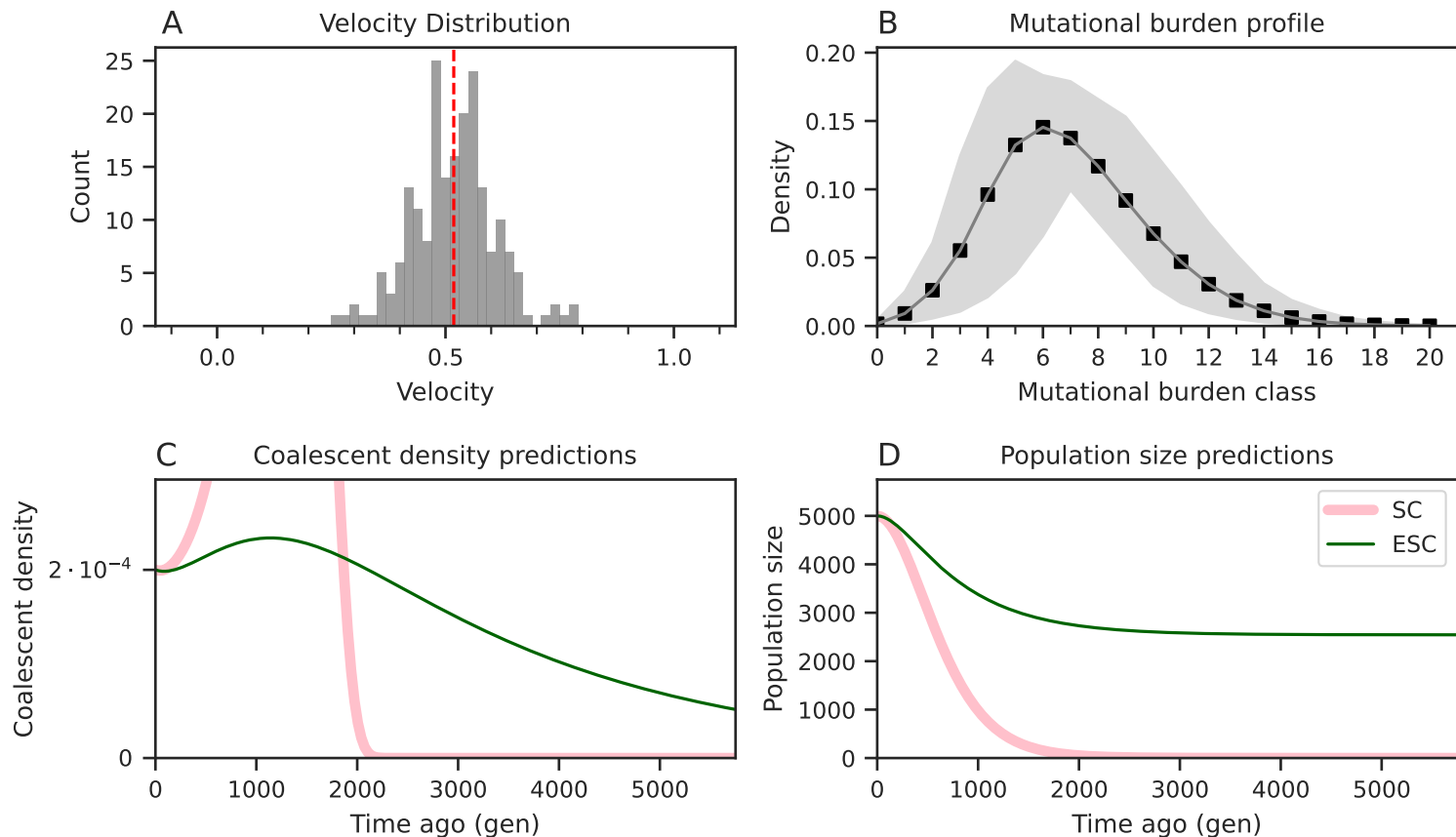
D Population size predictions



$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 1 \cdot 10^{-4}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5203 ± 0.0913 , $n = 200$ simulations



$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 5 \cdot 10^{-5}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5181 ± 0.0908 , $n = 200$ simulations



$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\sigma = 3 \cdot 10^{-5}$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity = 0.5199 ± 0.0856 , $n = 200$ simulations

